

Visualising High-dimensional Data with R

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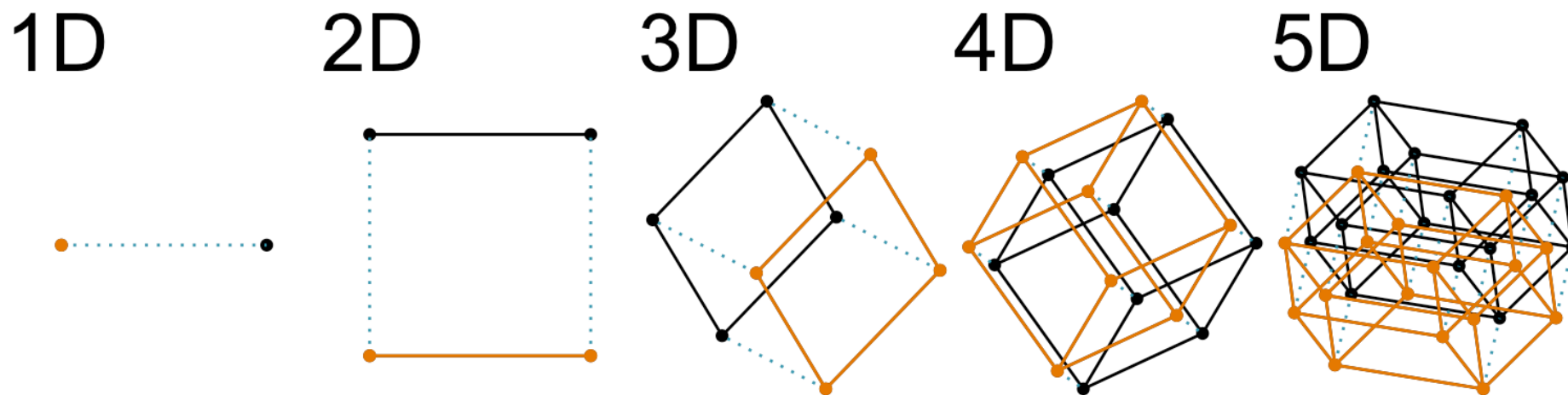
Session 1

Outline

time	topic
15	Introduction: What is high-dimensional data, why visualise and overview of methods
20	Basics of linear projections, and recognising high-d structure
25	Effectively reducing your data dimension, in association with non-linear dimension reduction

Introduction

What is high-dimensional space?



Increasing dimension adds an additional orthogonal axis.

If you want more high-dimensional shapes there is an R package, [geozoo](#), which will generate cubes, spheres, simplices, mobius strips, torii, boy surface, klein bottles, cones, various polytopes, ...

And read or watch [Flatland: A Romance of Many Dimensions \(1884\) Edwin Abbott](#).

Notation: Data

$$X_{n \times p} = [X_1 \ X_2 \ \dots \ X_p]_{n \times p} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}_{n \times p}$$

Notation: Projection

$$A_{p \times d} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1d} \\ a_{21} & a_{22} & \dots & a_{2d} \\ \vdots & \vdots & & \vdots \\ a_{p1} & a_{p2} & \dots & a_{pd} \end{bmatrix}_{p \times d}$$

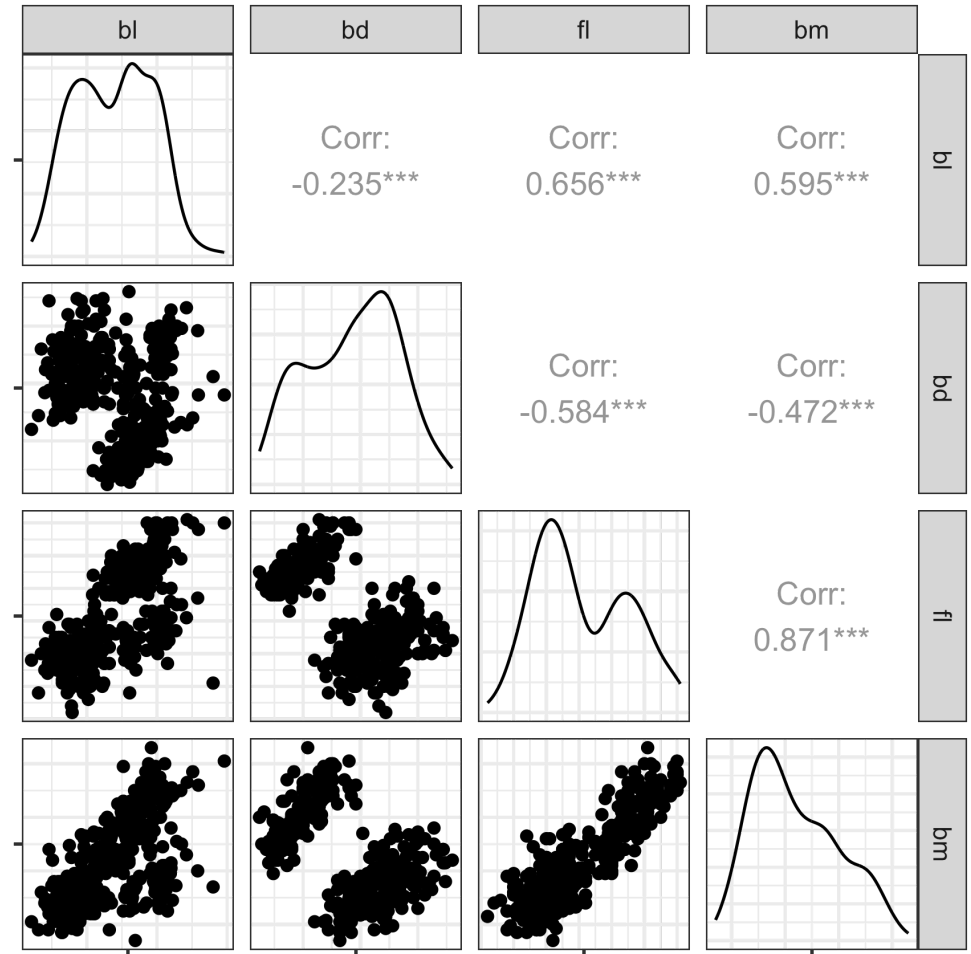
Notation: Projected data

$$Y_{n \times d} = XA = \begin{bmatrix} y_{11} & y_{12} & \dots & y_{1d} \\ y_{21} & y_{22} & \dots & y_{2d} \\ \vdots & \vdots & & \vdots \\ y_{n1} & y_{n2} & \dots & y_{nd} \end{bmatrix}_{n \times d}$$

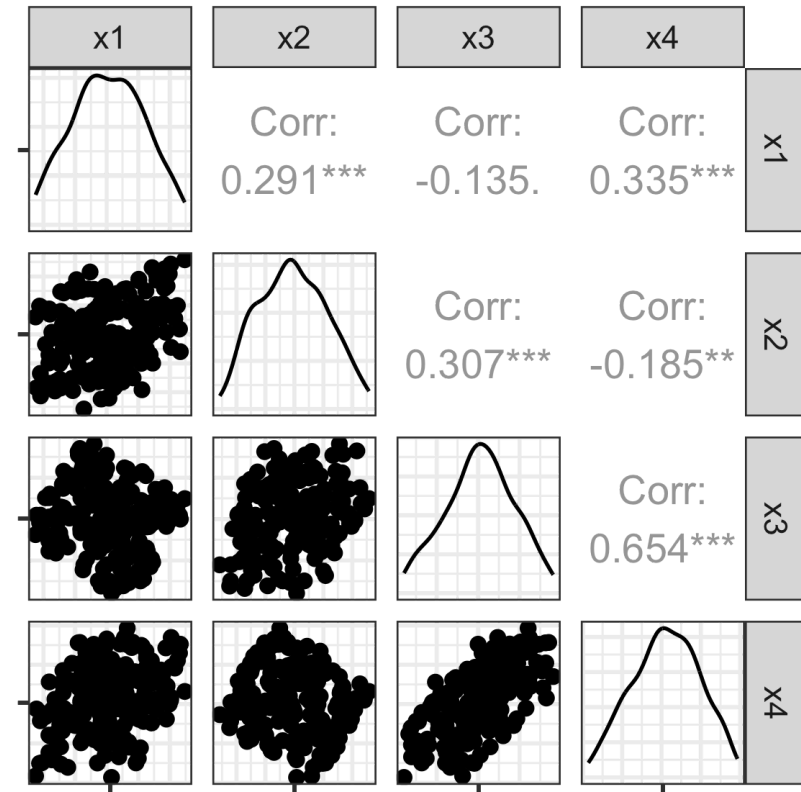
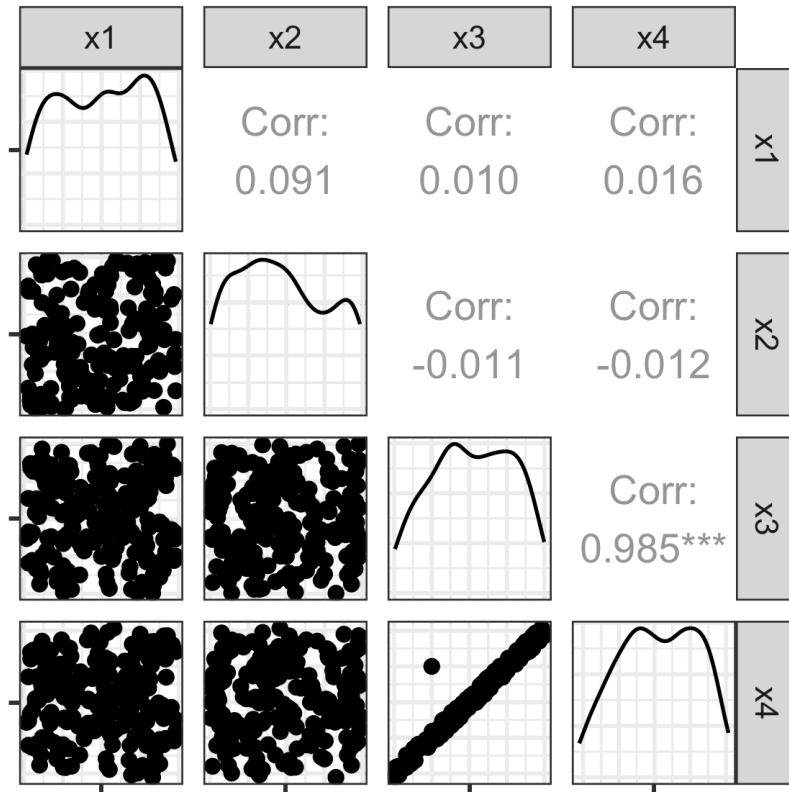
Why? (1/2)

Scatterplot matrix

Here, we see **linear association**, **clumping** and **clustering**, potentially some outliers.



Why? (2/2)



There is an outlier in the data on the right, like the one in the left, but it is **hidden in a combination of variables**. It's not visible in any pair of variables.

And help to see the data as a whole

To avoid misinterpretation see the bigger picture!

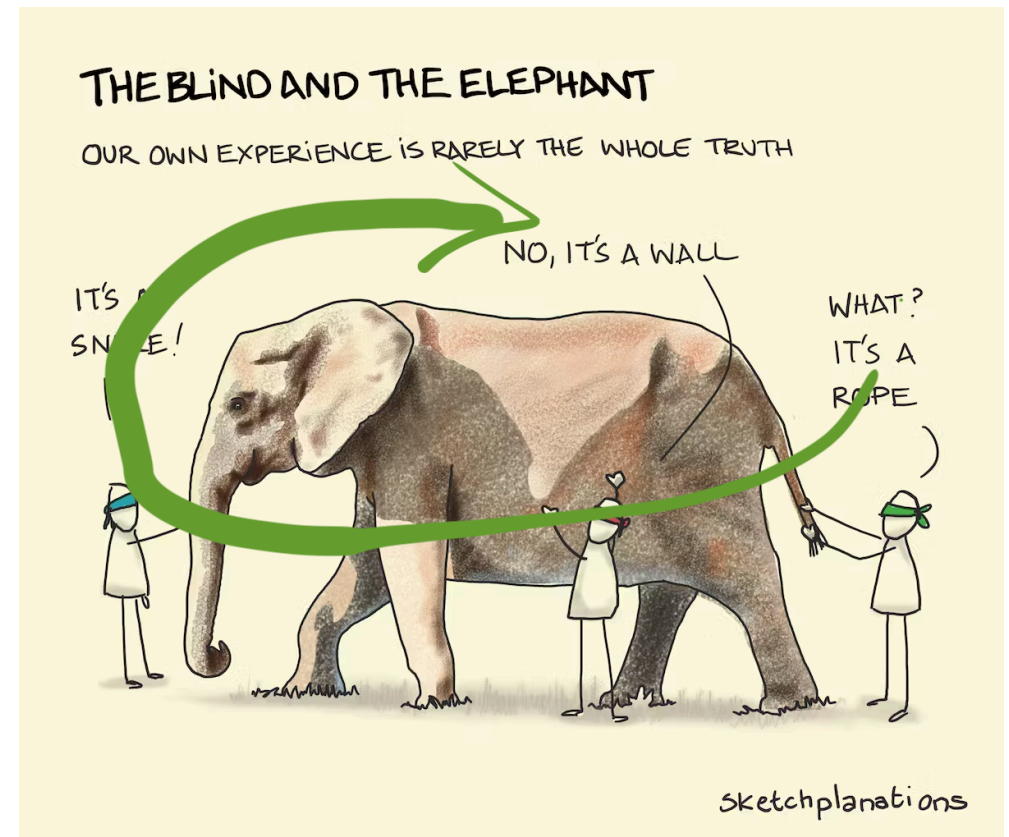
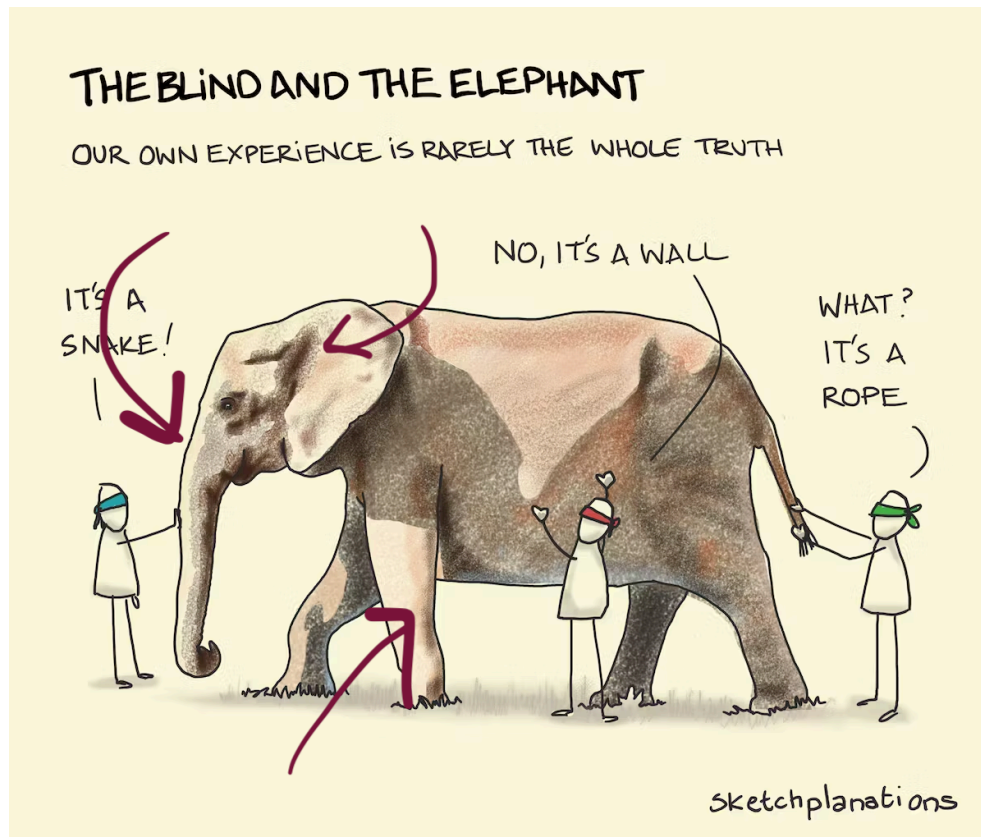
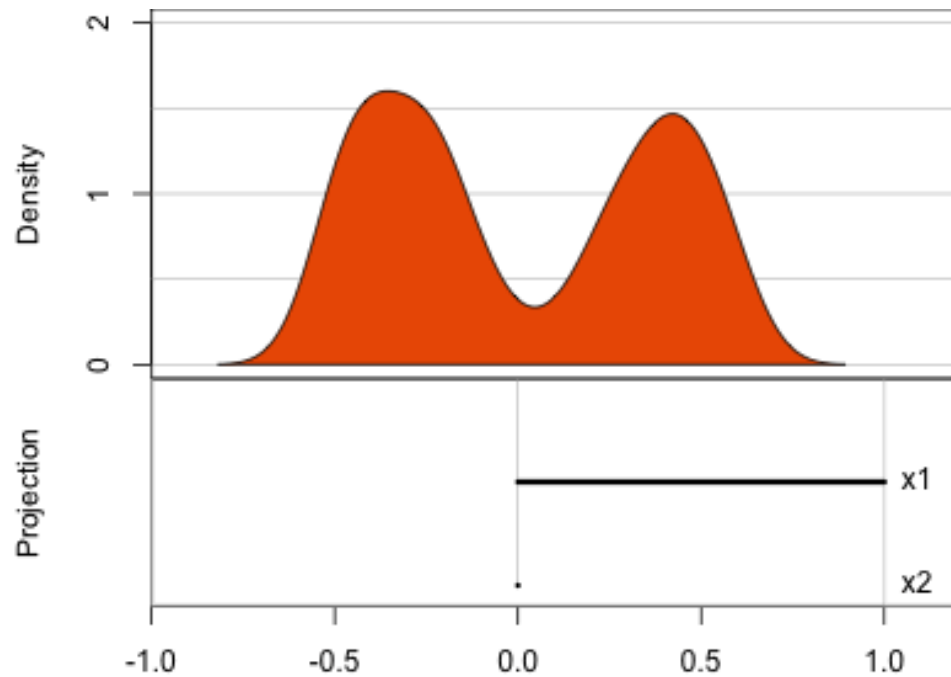


Image: [Sketchplanations](#).

<https://dicook.github.io/tutorial-highd-vis/>

Tours of linear projections

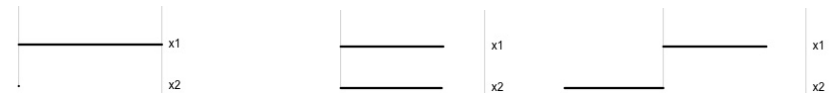


Data is 2D: $p = 2$

Projection is 1D: $d = 1$

$$A_{2 \times 1} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}_{2 \times 1}$$

Notice that the values of A change between $(-1, 1)$. All possible values being shown during the tour.



$$A = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0.7 \\ 0.7 \end{bmatrix}$$

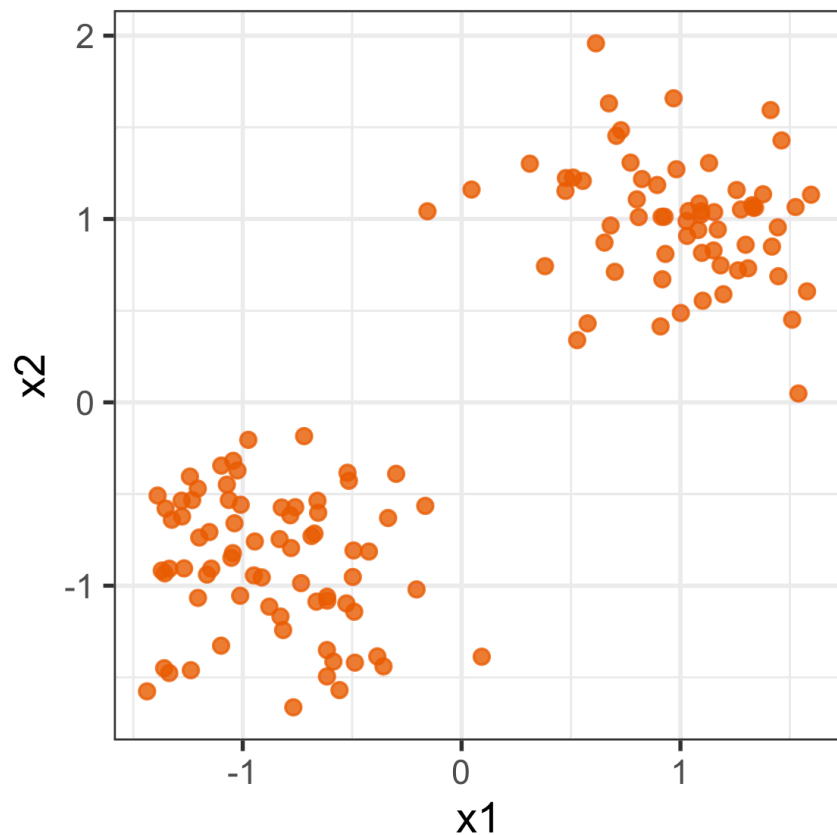
$$A = \begin{bmatrix} 0.7 \\ -0.7 \end{bmatrix}$$

watching the 1D shadows we can see:

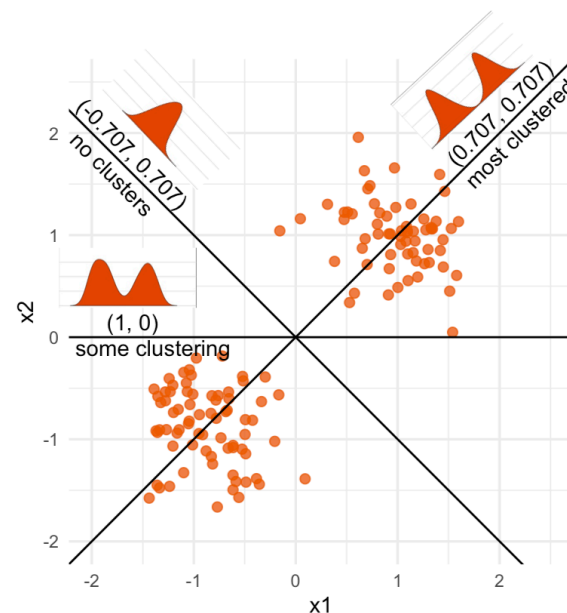
- unimodality
- bimodality, there are two clusters.

What does the 2D data look like? Can you sketch it?

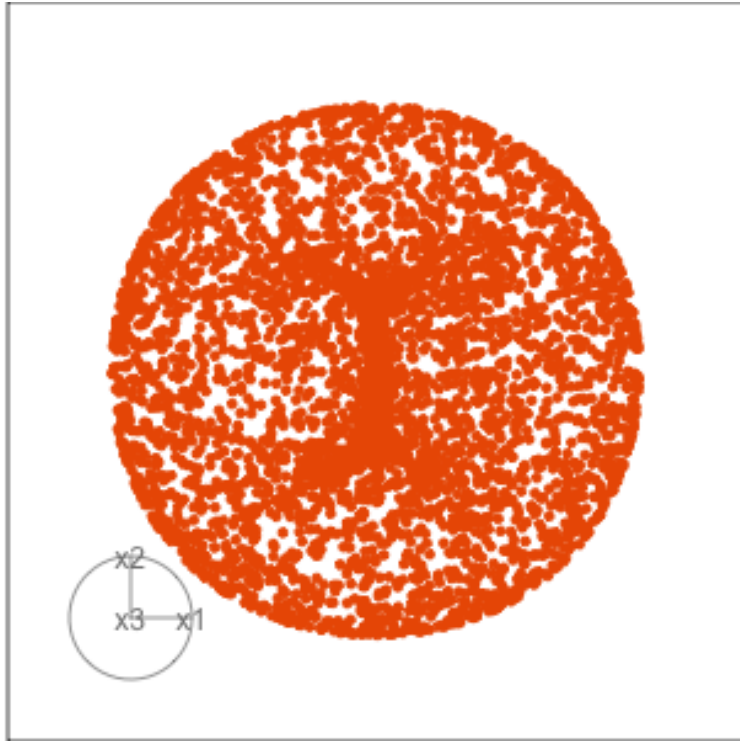
Tours of linear projections



←
The 2D data



Tours of linear projections



Data is 3D: $p = 3$

Projection is 2D: $d = 2$

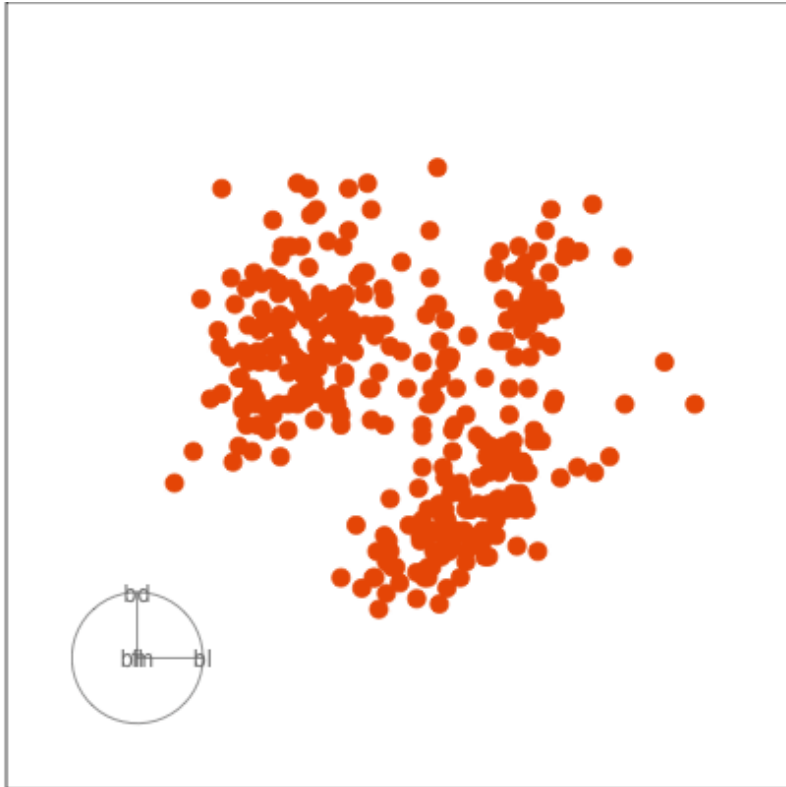
$$A_{3 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3 \times 2}$$

Notice that the values of A change between $(-1, 1)$. All possible values being shown during the tour.

See:

- circular shapes
- some transparency, reveals middle
- hole in in some projections
- no clustering

Tours of linear projections



Data is 4D: $p = 4$

Projection is 2D: $d = 2$

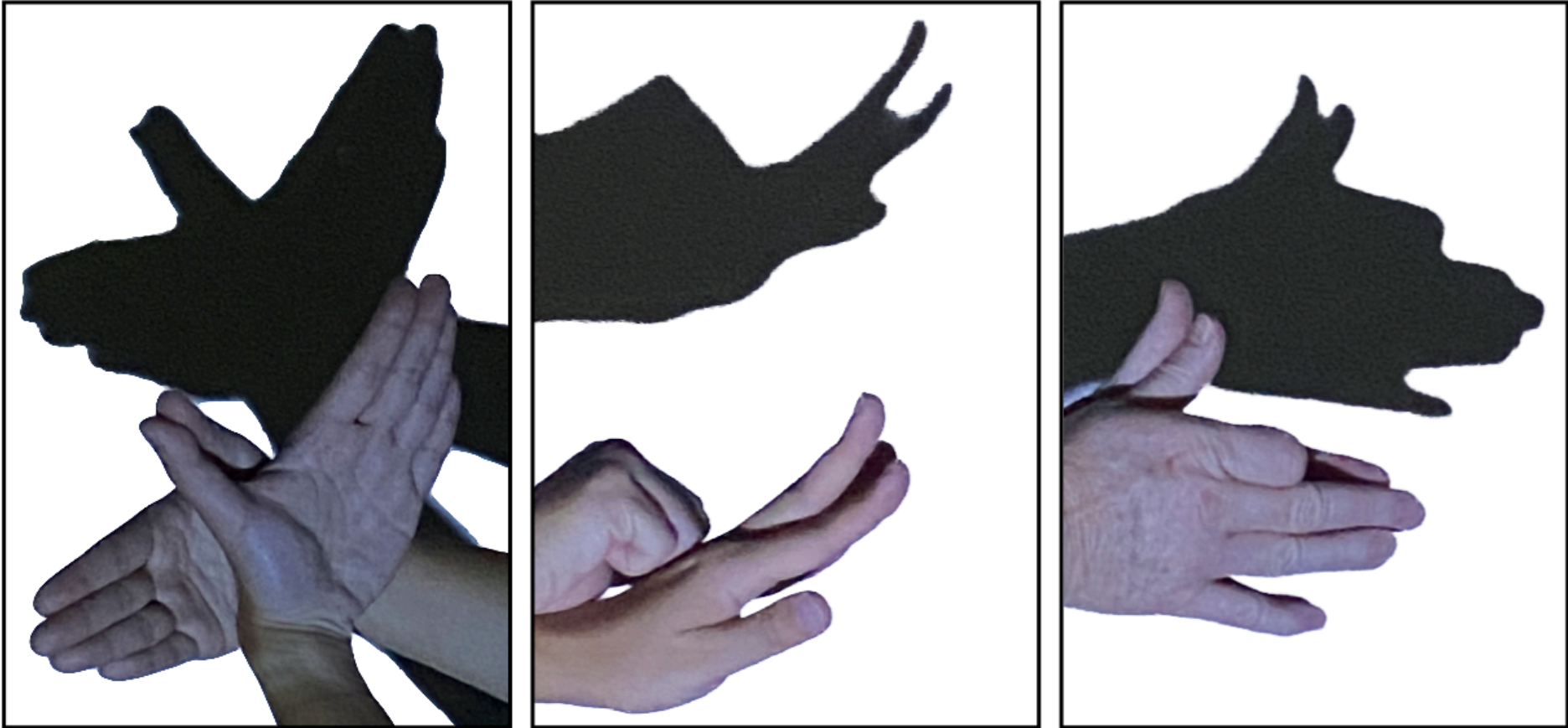
$$A_{4 \times 2} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \\ a_{41} & a_{42} \end{bmatrix}_{4 \times 2}$$

<https://dicook.github.io/tutorial-highd-vis/>

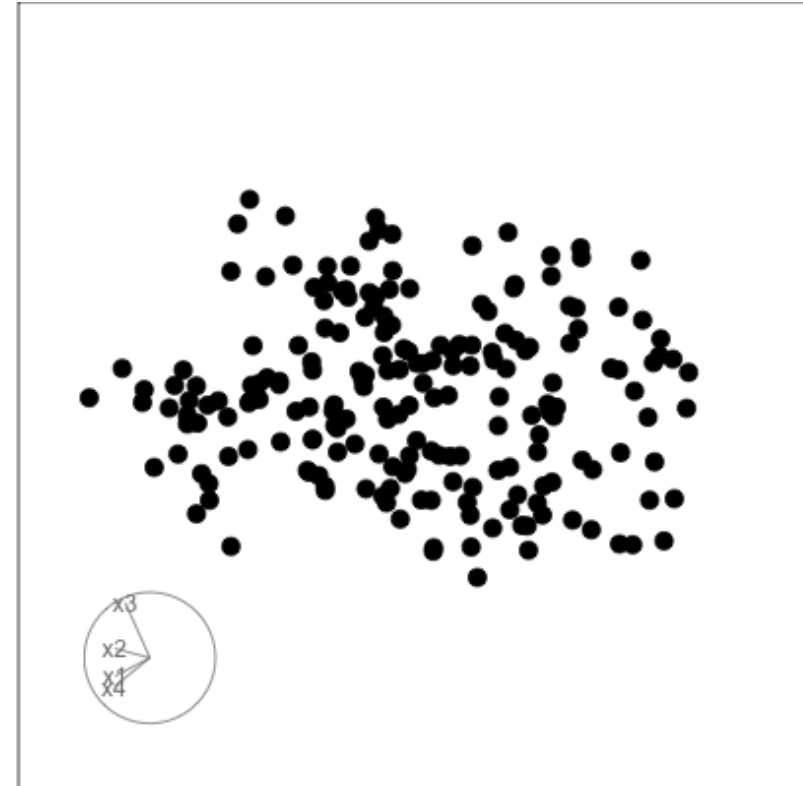
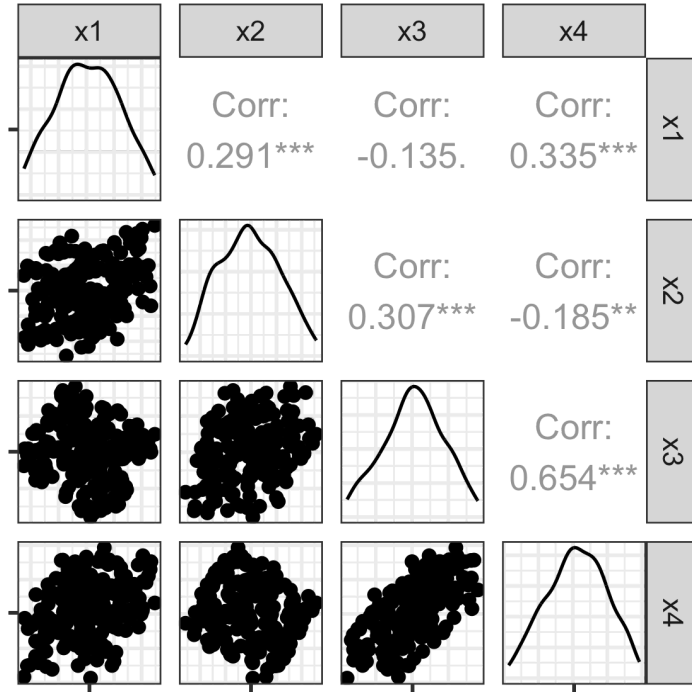
How many clusters do you see?

- three, right?
- one separated, and two very close,
- and they each have an elliptical shape.
- do you also see an outlier or two?

Intuitively, tours are like ...



Anomaly is no longer hidden



Wait for it!

How to use a tour in R

This is a **basic tour**, which will run in your plot window. (If this doesn't show, you can create a special plot window on OSX using `quartz()` or on Windows `X11()`.)

► Show code

This data has a class variable, **species**.

	species	tars1	tars2	head	aede1	aede2	aede3
1	Concinna	191	131	53	150	15	104
2	Concinna	185	134	50	147	13	105
3	Concinna	200	137	52	144	14	102

Use this to **colour points** with:

► Show code

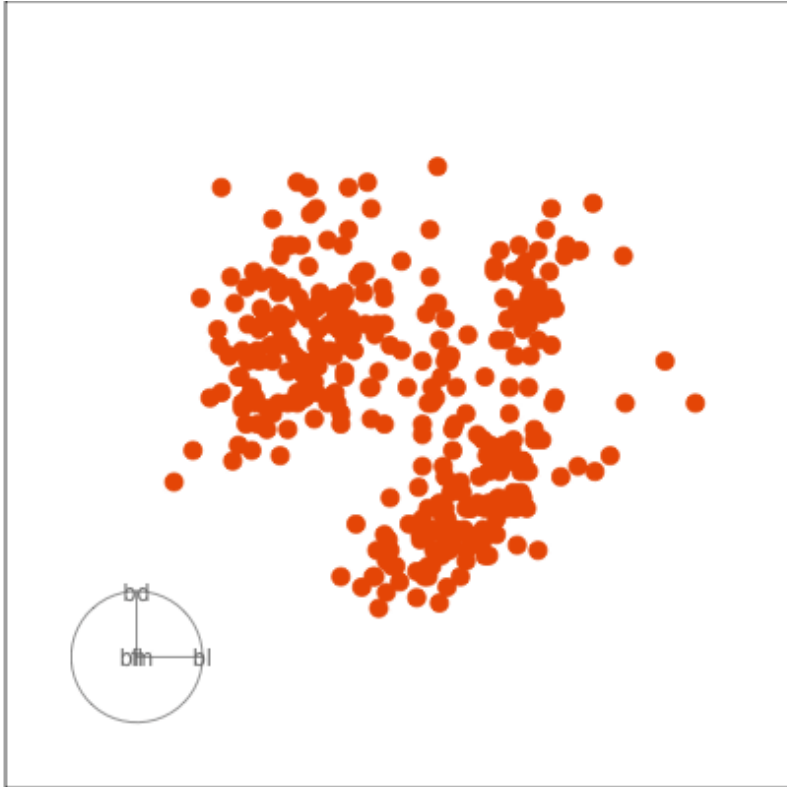
You can specifically **guide** the tour choice of projections using

► Show code

and you can **manually** choose a variable to control with:

► Show code

How to save a tour



To save as an animated gif:

► Show code

Your turn

Use a grand tour on the data set `c1` in the `mulgar` package.
What shapes do you see?

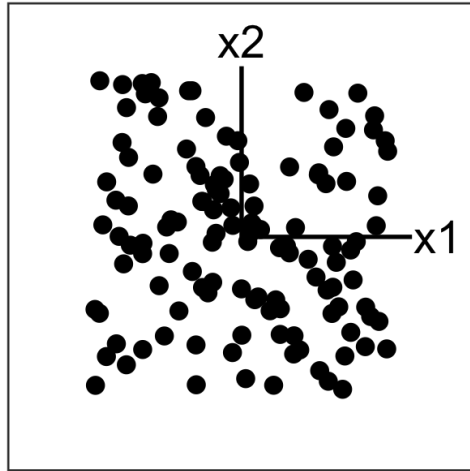
► Show code

Have a look at `c3` or `c7` also. How are the structures different.

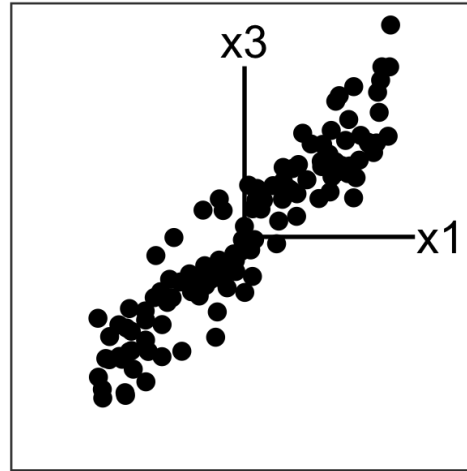
Dimension reduction

What is dimensionality?

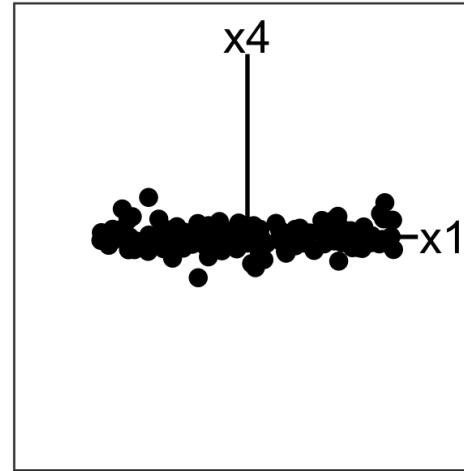
(a) Fully 2D



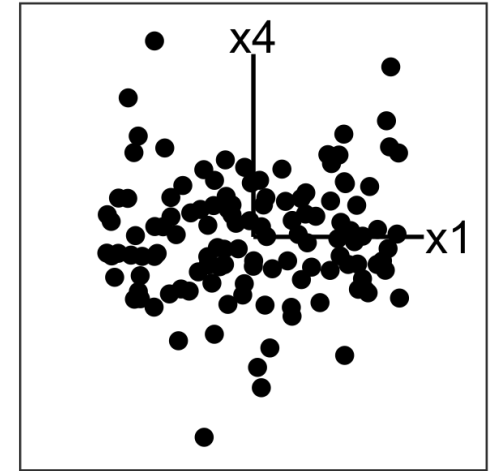
(b) Reduced dimension



(c) Reduced variance

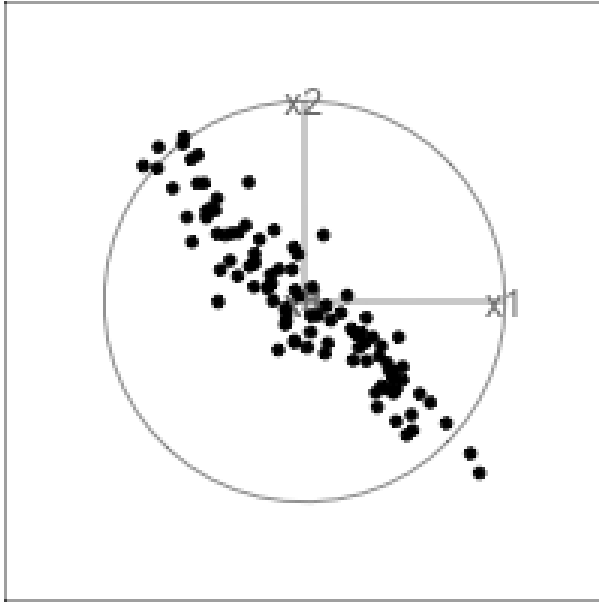


(d) Rescaled

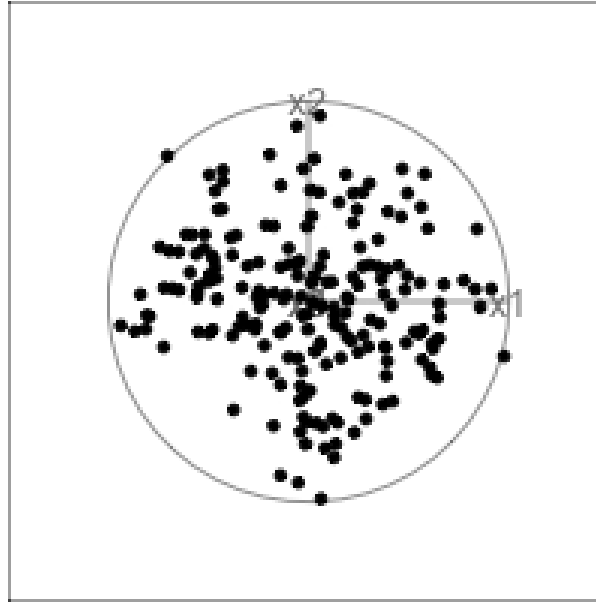


When an axis extends out of a direction where the points are collapsed, it means that this variable is partially responsible for the reduced dimension.

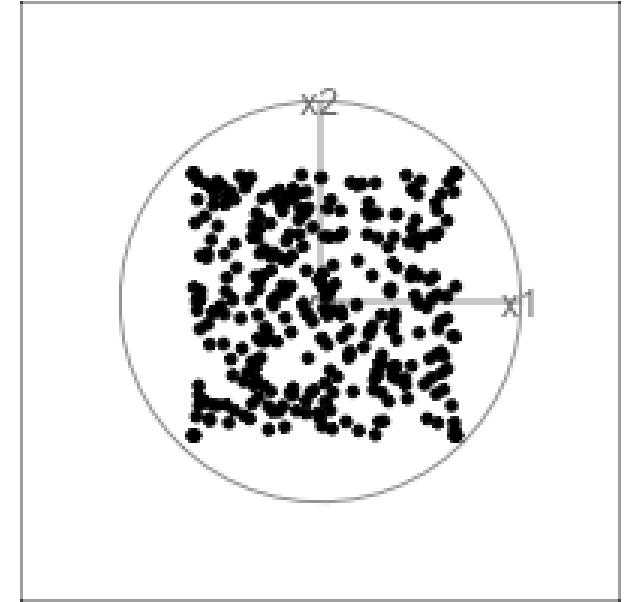
In high-dimensions



2D plane in 5D



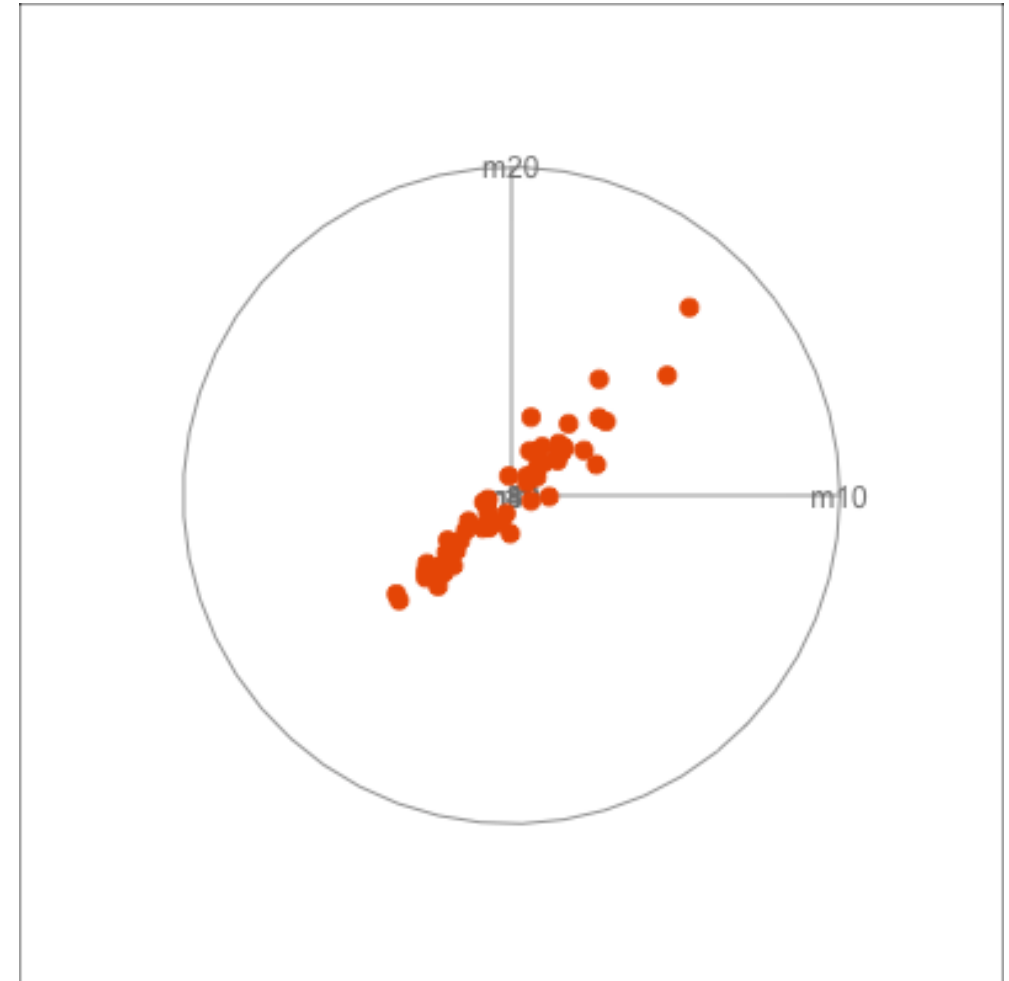
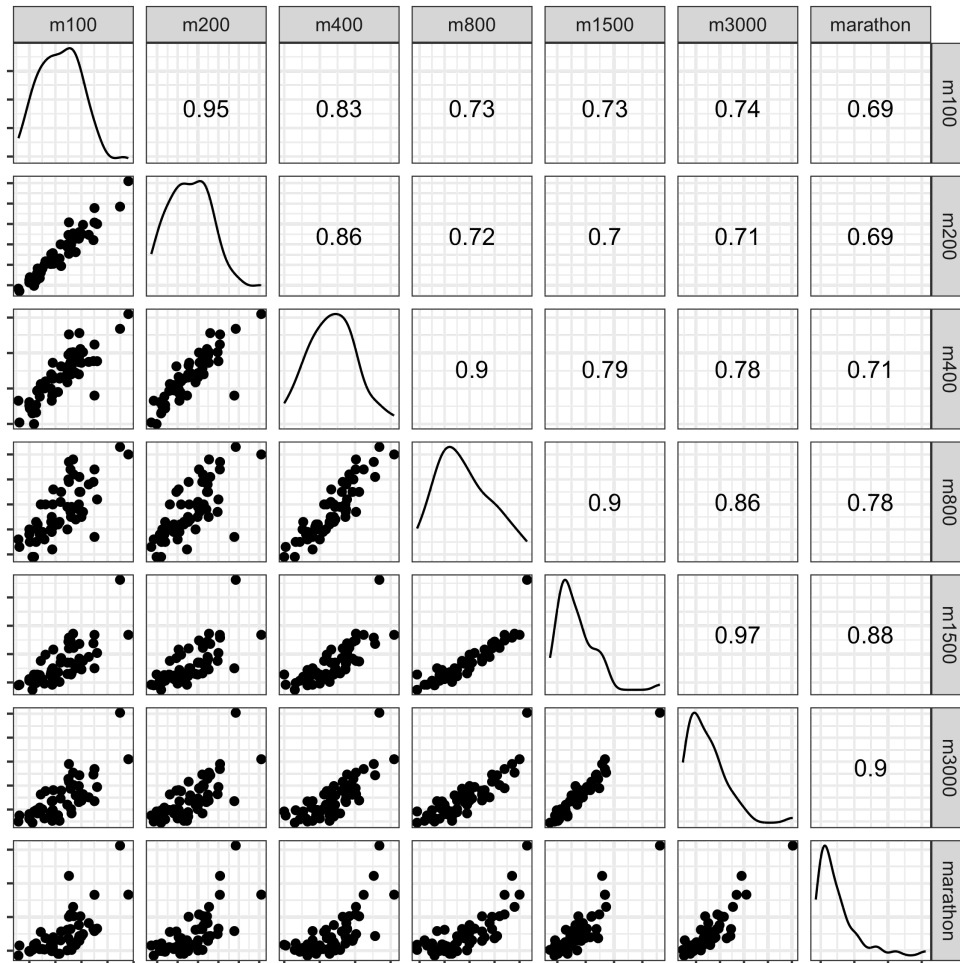
3D plane in 5D



5D plane in 5D

Principal component analysis (PCA) will detect these dimensionalities.

Example: womens' track records (1/3)



Example: PCA summary (2/3)

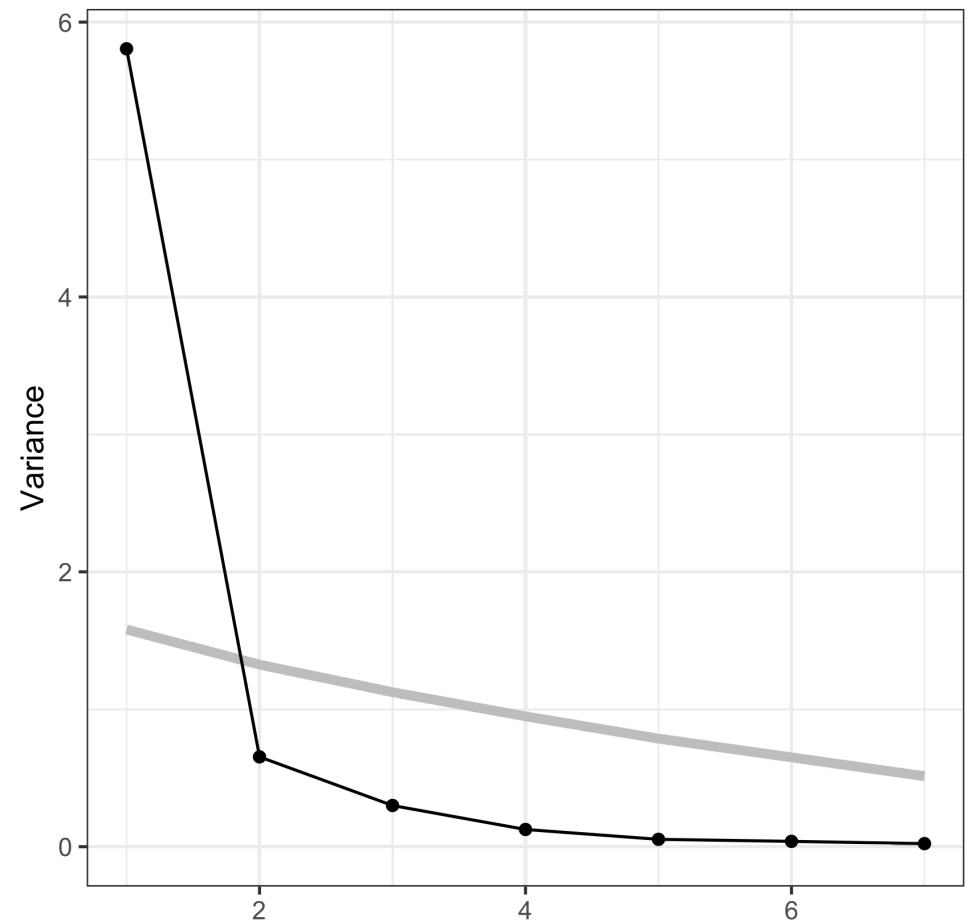
Variances/eigenvalues

```
[1] 5.806 0.654 0.300 0.125 0.054  
0.039 0.022
```

Component coefficients

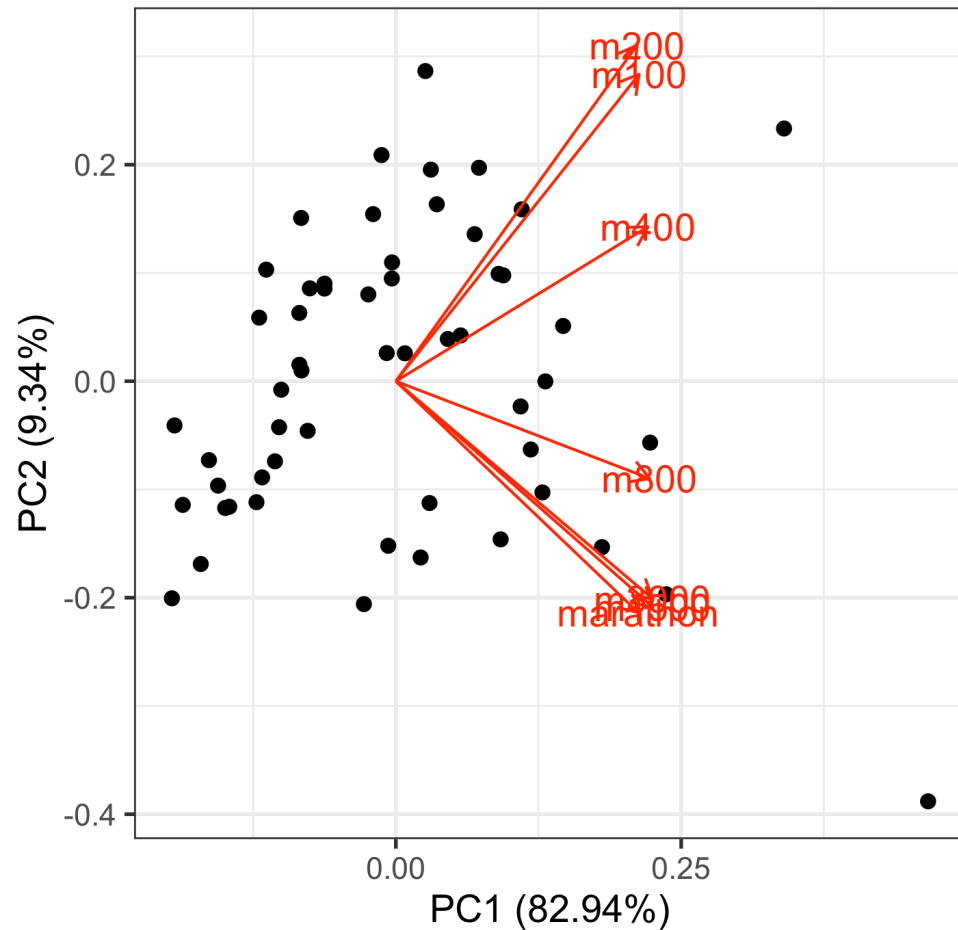
	PC1	PC2	PC3	PC4
m100	0.37	0.49	-0.286	0.319
m200	0.37	0.54	-0.230	-0.083
m400	0.38	0.25	0.515	-0.347
m800	0.38	-0.16	0.585	-0.042
m1500	0.39	-0.36	0.013	0.430
m3000	0.39	-0.35	-0.153	0.363
marathon	0.37	-0.37	-0.484	-0.672

How many PCs?

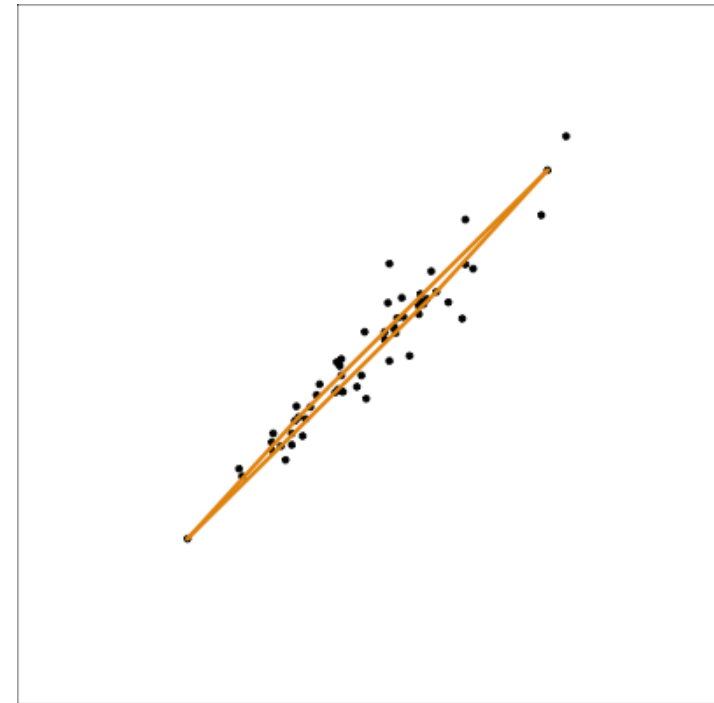


Example: Visualise (3/3)

Biplot: data in the model space



2D model in data space



► Show code

Non-linear dimension reduction (1/2)

Find some low-dimensional layout of points which approximates the distance between points in high-dimensions, with the purpose being to have a **useful representation that reveals high-dimensional patterns**, like clusters.

Multidimensional scaling (MDS) is the original approach:

$$\text{Stress}_D(x_1, \dots, x_n) = \left(\sum_{i,j=1; i \neq j}^n (d_{ij} - d_k(i, j))^2 \right)^{1/2}$$

where D is an $n \times n$ matrix of distances (d_{ij}) between all pairs of points, and $d_k(i, j)$ is the distance between the points in the low-dimensional space.

PCA is a special case of MDS. The result from PCA is a linear projection, but generally MDS can provide some non-linear transformation.

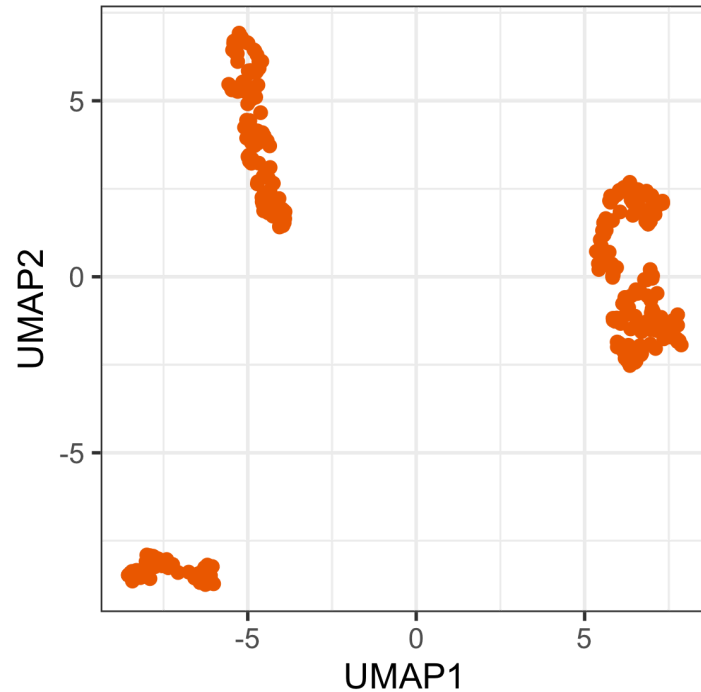
Many variations being developed:

- **t-stochastic neighbourhood embedding (t-SNE)**: compares interpoint distances with a standard probability distribution (eg t -distribution) to exaggerate local neighbourhood differences.
- **uniform manifold approximation and projection (UMAP)**: compares the interpoint distances with what might be expected if the data was uniformly distributed in the high-dimensions.

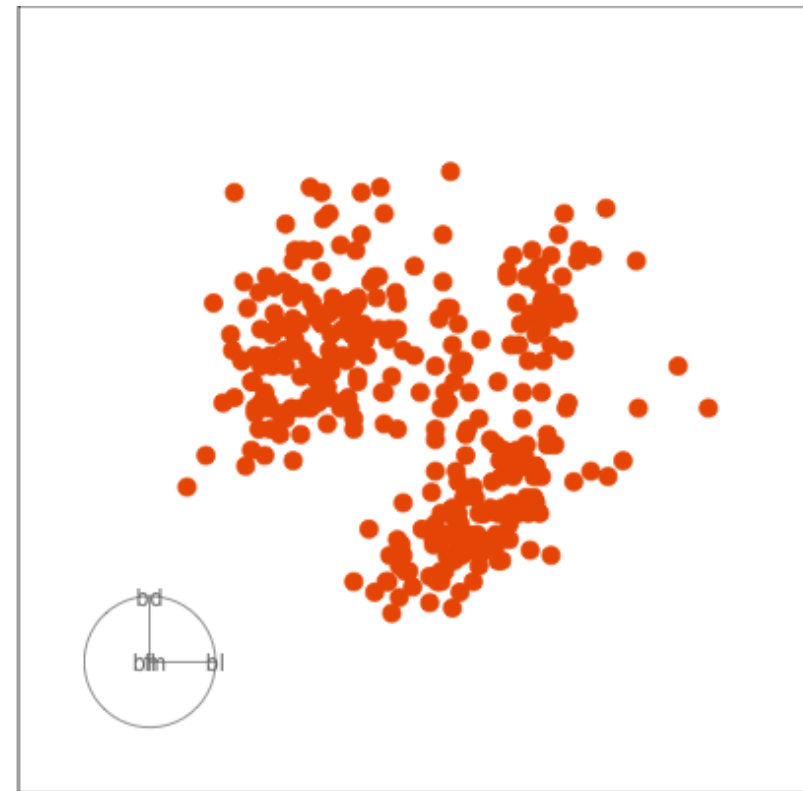
NLDR can be useful but it can also make some misleading representations.

Non-linear dimension reduction (2/2)

UMAP 2D representation



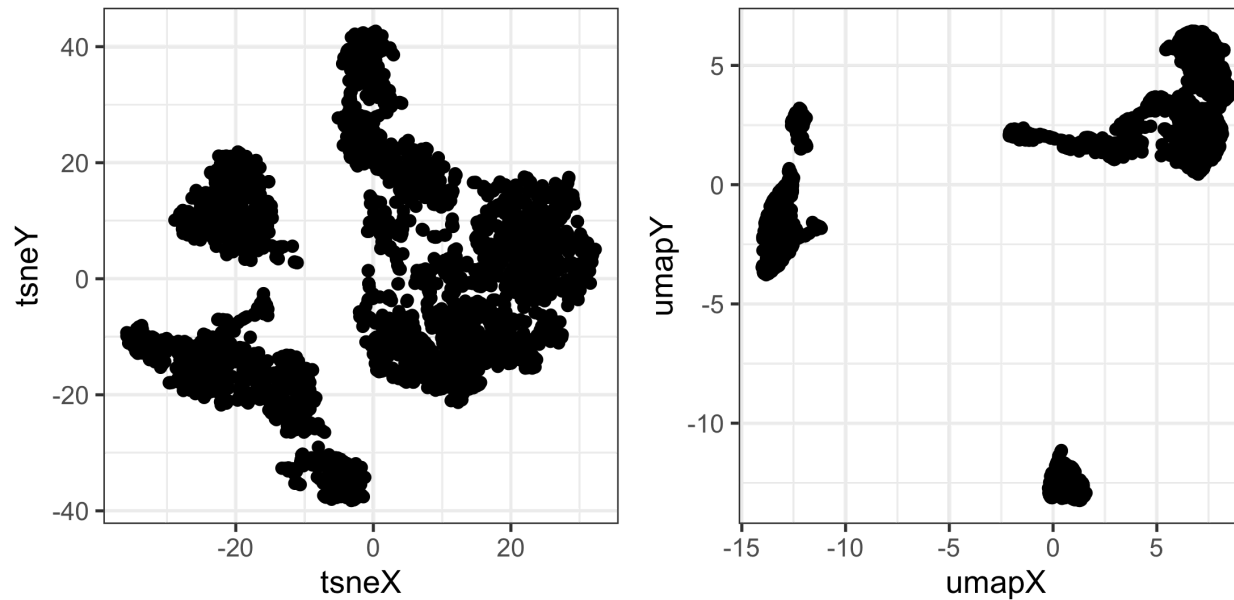
Tour animation of the same data



► Show code

Your turn

Which is the best representation, t-SNE or UMAP, of this 9D data?



You can use this code to read the data and view in a tour:

► [Show code](#)

Key conceptual points

- Avoid misinterpretation, by using your high-dimensional visualisation skills to look at the **data as a whole**.
- Examine model fit by examining the model overlaid on the data, **model-in-the-data-space**. ([Wickham et al \(2015\) Removing the Blindfold](#))

End of session 1



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